



Layer 2 Switch Attacks and Mitigations for ARP Spoofing, MAC Overflow and DHCP Starvation attacks

Krish Thapaliya

Purpose

This guide/analysis is meant for personnel who wish to learn 3 methods on how to hack another network device on a network as well as how to preemptively mitigate these attacks from happening. These 3 methods are the DHCP Starvation attack, MAC Flooding attack and an ARP Spoofing attack. This guide/analysis for these 3 methods will include multiple steps that the average personnel would not be familiar with such as downloading Linux on the same computer that is running Windows, Using a tool called Yersinia and being able to analyze network traffic in order to gain greater insight into a network with the application Wireshark.

Background

There are 3 different attacks that this guide/analysis will be explaining on how to both conduct and mitigate in a real applicable network. These 3 attacks are a DHCP Starvation Attack, ARP Spoofing and MAC Flooding. The core trait that these attacks have is that they are all categorized as a Layer 2 switch attack. Layer 2 attacks distinguish themselves by targeting only the Data Link Layer, which is Layer 2 in the OSI model, to manipulate local network traffic by exploiting vulnerabilities in MAC address tables, ARP and DHCP protocols. These attacks happen within the same broadcast domain allowing attacks to intercept data or bypass certain restrictions set, without needing a router.

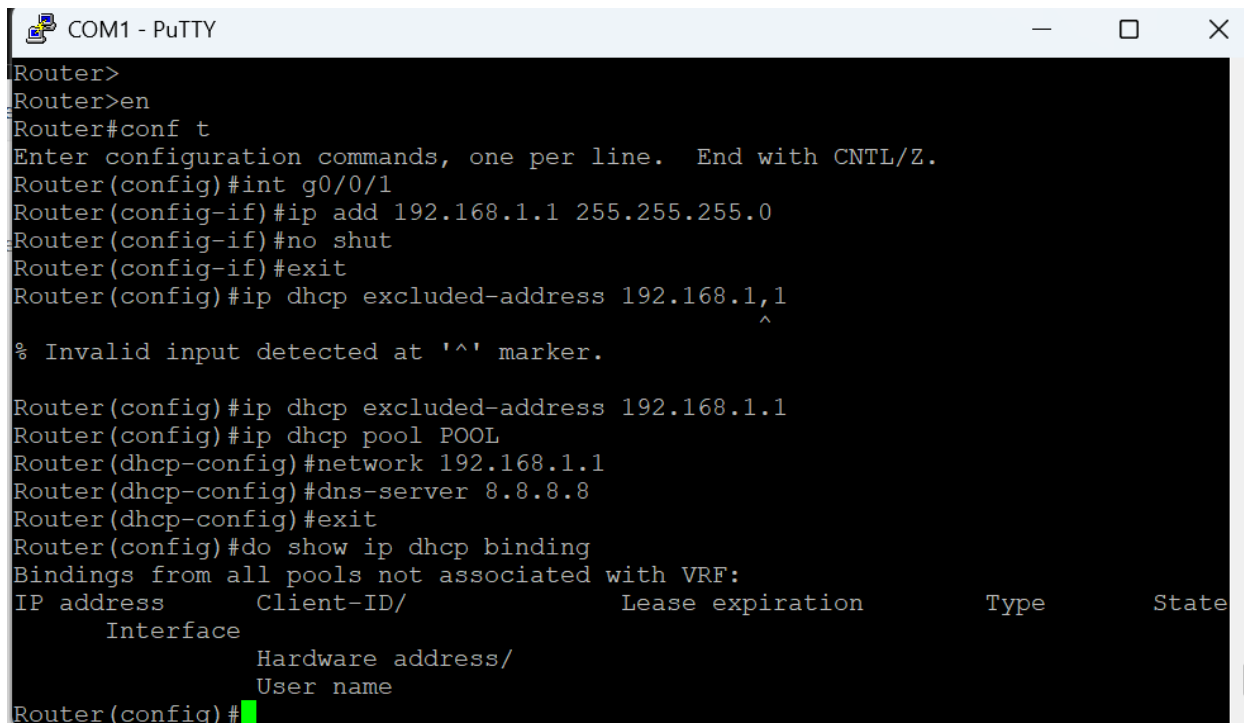
The first attack that is explained in this guide is the DHCP Starvation attack. This is an attack that is conducted on a computer network, where the entire possible range of available IP addresses assigned by DHCP is consumed/filled up. The reason that having the entire DHCP IP range filled up is a problem is due to the nature of how Automatic DHCP works on computer networks. When any new device is joining a network, the device itself is assigned an IP address for identification purposes, however with this attack the network will not be able to automatically assign any more IP addresses to new devices essentially blocking any new devices to have access to the network and if any existing devices leave the network while the attack is being conducted, those devices as well will not be able to rejoin the network. The reason that this attack is able to work on a network is because this attack exploits the fact that a MAC address is now changed from an address that is permanently fixed to the hardware to freely configured on many network interfaces. The DHCP Attack itself are thousands of unique MAC addresses that are used to identify devices and are used to trick the server or in this case the switch into believing that there thousands of legitimate devices that are requesting IP addresses to join the network and causing the server or in this case the switch to completely use up all the IP addresses.

The second attack that is explained in this guide is the ARP Spoofing attack. This is an attack that is conducted on a computer network, where an attacker sends an Address Resolution Protocol (ARP) messages onto a local area network. This will make the network associate the attacker's MAC address with the IP address of another host making any traffic meant for that IP address to be sent to the attacker instead. ARP spoofing also allows an attack to manipulate all the traffic on the network and is used as an opening attack in a black hat's reservoir of tools to easily conduct more dangerous attacks on a network. However, this attack can be conducted on a network that uses the Address Resolution Protocol (ARP) and the attacker should have direct access to the local network segment that they want to attack for the minimum requirements of allowing the attack to be run successfully. The Address Resolution Protocol normally functions by first having a network device sending a broadcast packet throughout the network which is known as an ARP request. The second network device with the IP address listed in the ARP request will then respond with a broadcast ARP reply that is spread through the entire network which contains the MAC address associated with its IP address. ARP itself is a stateless protocol which means that the receiver of the protocol does not store the session state from previous requests but instead transfers the relevant session state to the receiver in such a way that every request can be understood in isolation, without reference to the session state from the previous request. There is no method in the ARP protocol by which a host can authenticate the peer from which the packet originated. This method is the vulnerability that allows ARP spoofing to be successful.

The third attack that is explained in this guide is the MAC Flooding Attack. In networking, a Media Access Control (MAC) flooding is a technique that is used to compromise the security of specifically network switches. This attack works by extracting legitimate MAC table contents out of the switch and forcing unicast flooding behavior which means sending information throughout the network from all ports of the network device causing potentially sensitive information to be sent to portions of the network where it is not normally intended to go. Switches are a type of network device, will keep a MAC table that stores individual MAC addresses on the network to the physical ports on the switch. This allows the switch to control where to send data out of the physical port where the recipient is located instead of just broadcasting the data out of all ports throughout the network providing vulnerabilities that people with malicious intentions can take advantage of. The advantage of this technique is that data is bridged exclusively to the network segment containing the computer that the data is specifically marked out for. In a normal Mac flooding attack a switch receives many Ethernet frames each containing a different source MAC address, by the attacker. The intention is to use up all the limited memory set aside in

the switch for the MAC address table. The desired effect is to essentially force legitimate MAC addresses out of the MAC address table using the facts that the attacker can send as many different source MAC addresses out on a network and the switch can only store a limited amount of source MAC addresses causing large amounts of incoming frames to be flooded out on all ports. Then after launching the successful MAC flooding attack, a malicious user can use a packet analyzer to get access to sensitive data being sent between other network devices like computers which would not normally be accessible if the switch was functioning normally. The attacker could also continue with an ARP spoofing attack which would allow them to keep access to the sensitive data even after the switches recover from the initial MAC flooding attack.

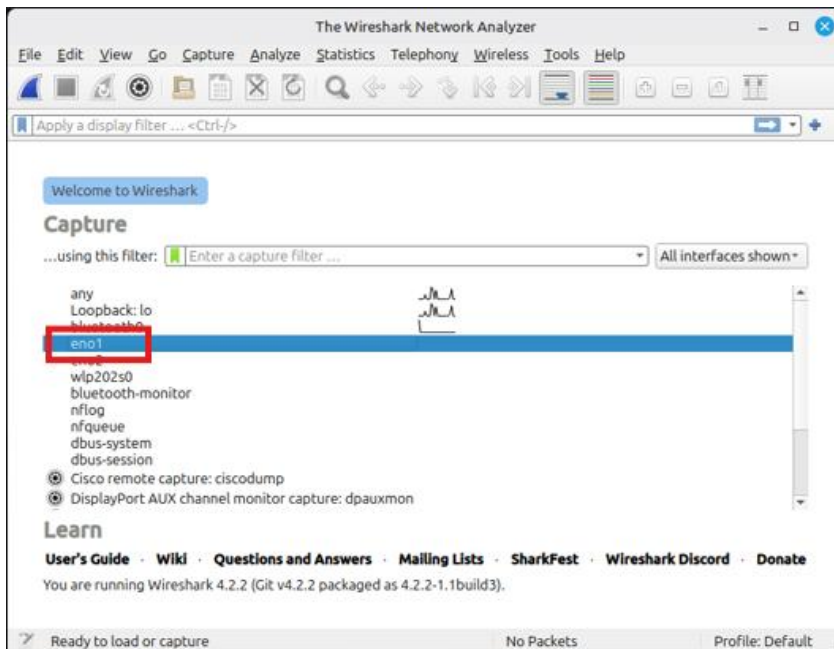
Steps:



```
Router>
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int g0/0/1
Router(config-if)#ip add 192.168.1.1 255.255.255.0
Router(config-if)#no shut
Router(config-if)#exit
Router(config)#ip dhcp excluded-address 192.168.1,1
^
% Invalid input detected at '^' marker.

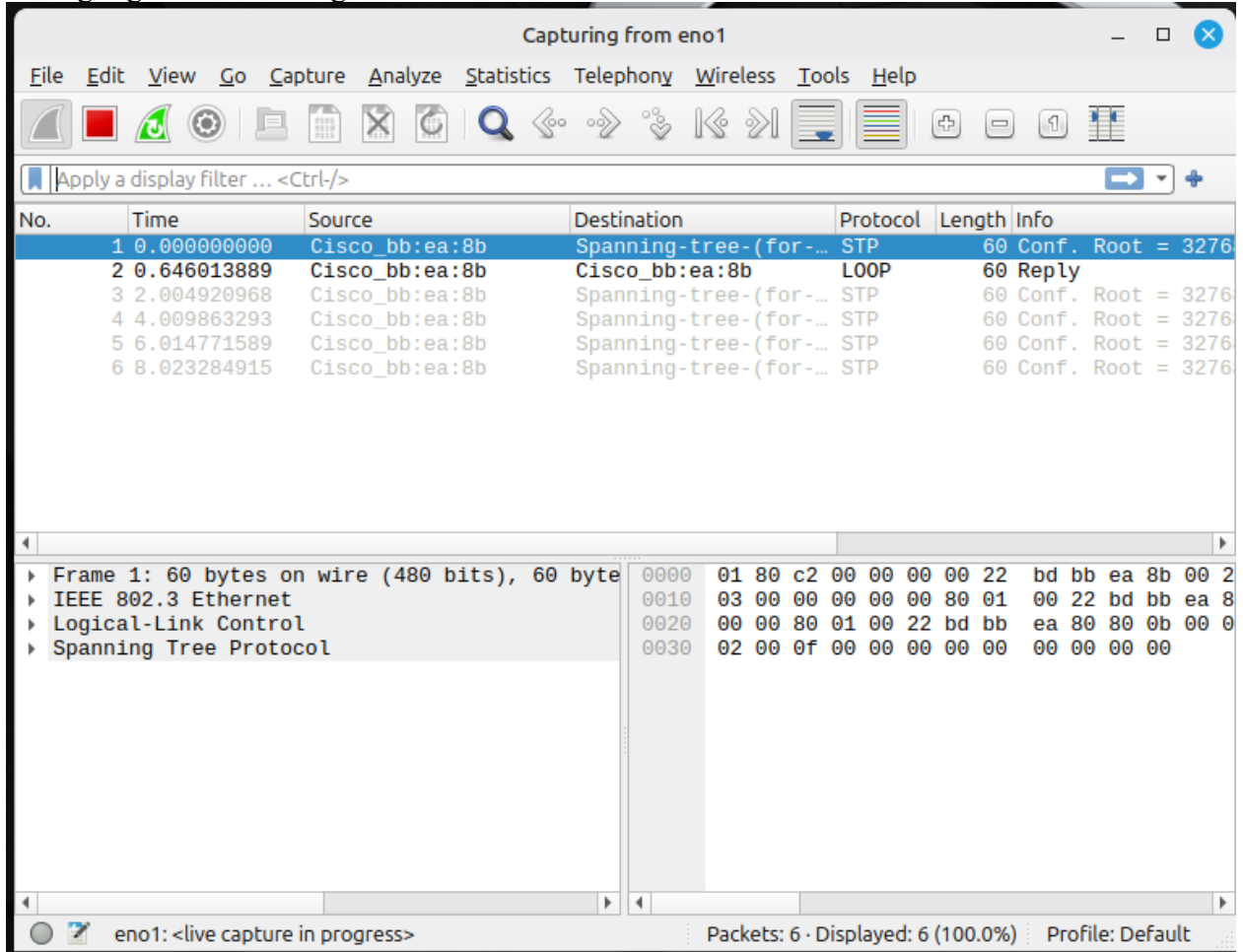
Router(config)#ip dhcp excluded-address 192.168.1.1
Router(config)#ip dhcp pool POOL
Router(dhcp-config)#network 192.168.1.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#do show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration        Type      State
  Interface
      Hardware address/
      User name
Router(config)#
```

1. Type the following commands as written in the image above in order to set up the network for basic DHCP in order to do a DHCP starvation attack.

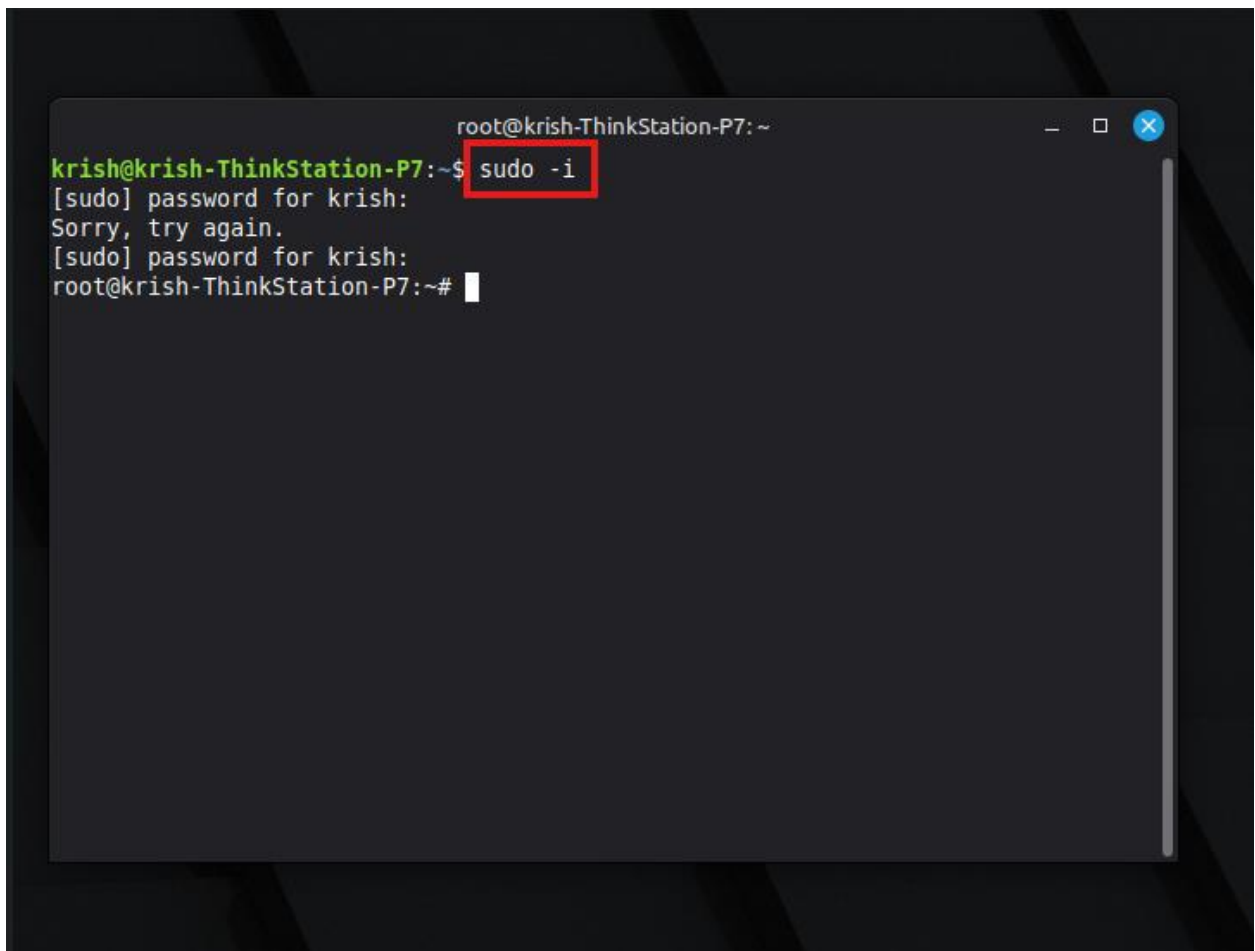


```
krish@krish-ThinkStation-P7:~  
krish@krish-ThinkStation-P7:~$ sudo wireshark  
[sudo] password for krish:  
** (wireshark:4164) 07:52:55.009265 [GUI WARNING] -- QStandardPaths: XDG_RUNTIME  
DIR not set, defaulting to '/tmp/runtime-root'  
□
```

2. Type the `sudo wireshark` command in terminal and select the eno1 interface as highlighted in the image above.

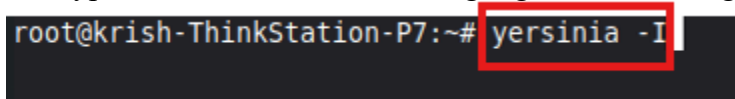


3. The wireshark application should then an interface similar to the image above

A terminal window titled "root@krish-ThinkStation-P7: ~" with standard window controls. The prompt is "krish@krish-ThinkStation-P7:~\$". The command "sudo -i" is entered and highlighted with a red box. The terminal shows the password prompt "[sudo] password for krish:", an error message "Sorry, try again.", and another password prompt "[sudo] password for krish:". After successful authentication, the prompt changes to "root@krish-ThinkStation-P7:~#" and a cursor is visible at the end of the line.

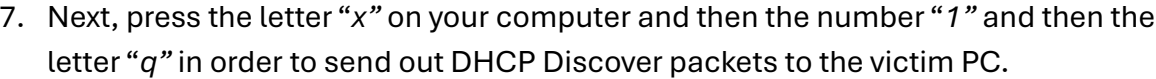
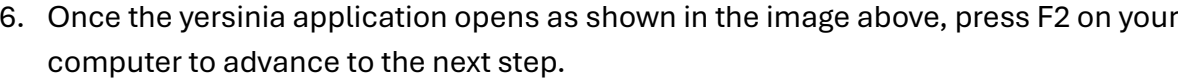
```
root@krish-ThinkStation-P7: ~
krish@krish-ThinkStation-P7:~$ sudo -i
[sudo] password for krish:
Sorry, try again.
[sudo] password for krish:
root@krish-ThinkStation-P7:~#
```

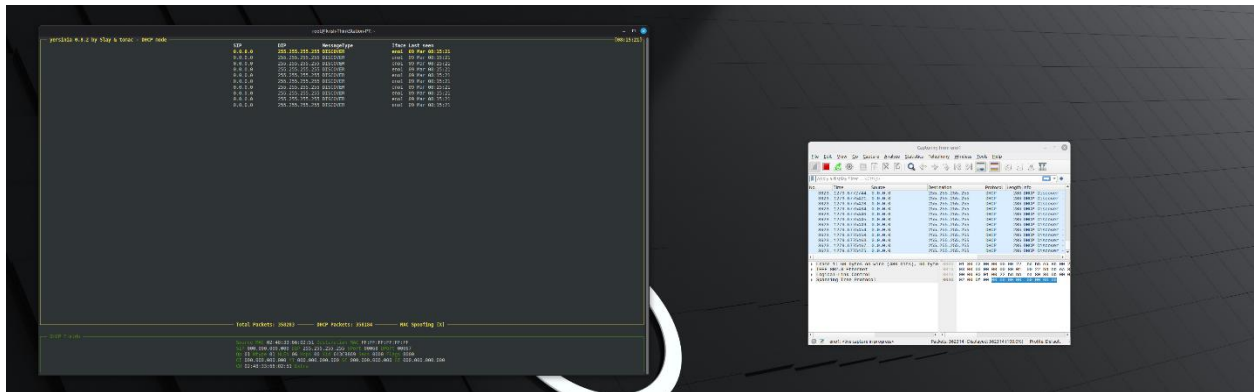
4. Type the *sudo -i* command as highlighted in the image above.

A terminal window showing the prompt "root@krish-ThinkStation-P7:~#". The command "yersinia -I" is entered and highlighted with a red box. A cursor is visible at the end of the command.

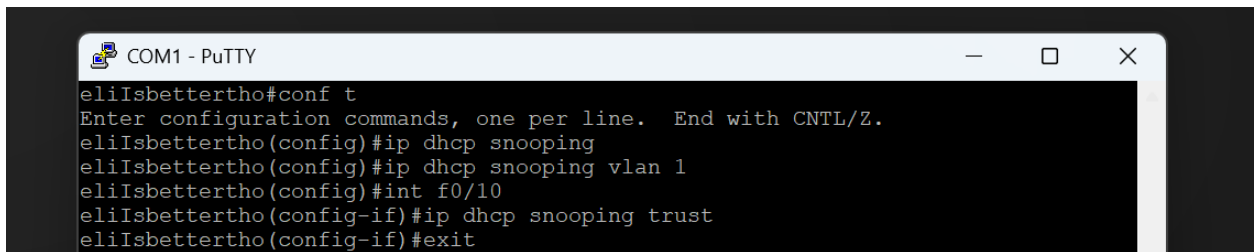
```
root@krish-ThinkStation-P7:~# yersinia -I
```

5. Type the *yersinia -I* command as highlighted in the image.





8. After you complete step 7, the Wireshark application should be constantly showing DHCP Discover packets and the yersinia application should be showing the attack working as well as the time the attack started. Next, I will be showing you how to mitigate a DHCP starvation attack.



```
eliIsbettertho(config)#int ra fa0/1-9 , fa0/11 - 24
eliIsbettertho(config-if-range)#ip dhcp snooping limit rate 5
eliIsbettertho(config-if-range)#switchport mode acc
eliIsbettertho(config-if-range)#switchport port-security
eliIsbettertho(config-if-range)#
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#do clear ip dhcp binding
% Incomplete command.

Router(config)#do clear ip dhcp binding ?
*          Clear all automatic bindings
A.B.C.D    Clear a specific binding
vrf        DHCP vrf bindings

Router(config)#do clear ip dhcp binding *
Router(config)#
```

9. On the Switch (name: eliIsbettertho) and Router type all of the following commands in both of the images above to prevent a DHCP Starvation attack from occurring on your network.

ARP Spoofing:

Note: This guide to ARP Spoofing is applicable either a virtual machine that is virtually running kali Linux or a partitioned drive that has kali Linux permanently

```
Ethernet adapter Ethernet:

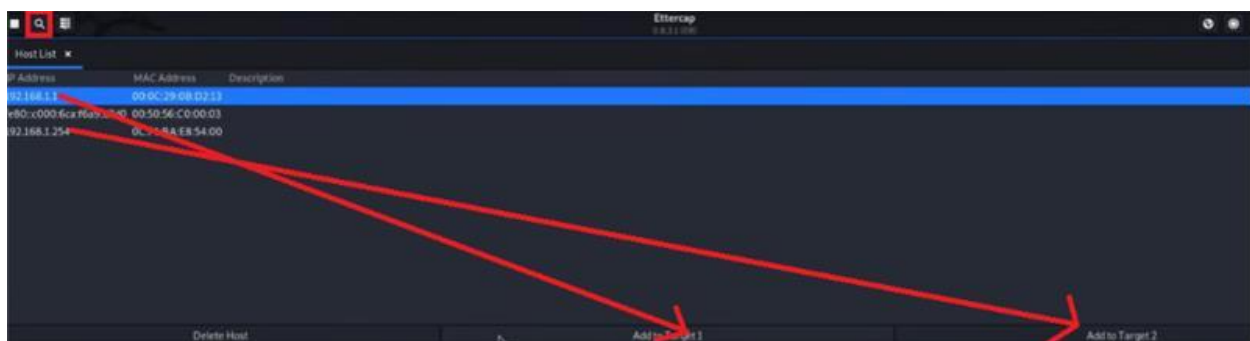
Connection-specific DNS Suffix . . : 
Description . . . . . : Realtek PCIe 2.5GbE Family Controller
Physical Address. . . . . : 58-11-22-48-E0-8D
DHCP Enabled. . . . . : yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::e3b0:4d5c:12a6:c200%12(Preferred)
IPv4 Address. . . . . : 192.168.221.116(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Saturday, March 7, 2026 12:48:31 AM
Lease Expires . . . . . : Saturday, March 14, 2026 12:48:28 AM
Default Gateway . . . . . : 192.168.221.99
DHCP Server . . . . . : 192.168.221.99
DHCPv6 IAID . . . . . : 106434850
DHCPv6 Client DUID. . . . . : 00-01-00-01-2A-C1-67-92-58-11-22-48-E0-8D
DNS Servers . . . . . : 75.75.75.75
                          75.75.76.76
NetBIOS over Tcpip. . . . . : Enabled

C:\Users\Luke>
```

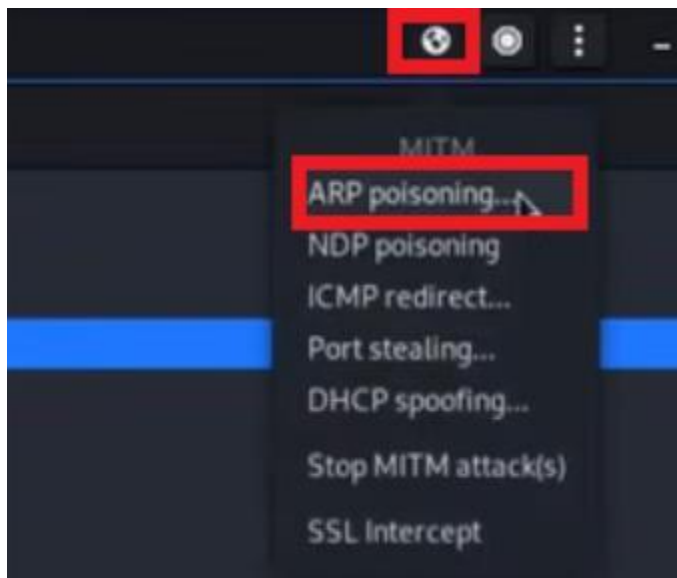
1. Type in the command “ipconfig/ all” if on Windows or “ip a” if on Linux and then locate your ip address and MAC address (physical address) as highlighted in the image above. Make sure to take note of these as they will be used in a later step.
2. First make sure you are in Linux, and then navigate to the tools tab and search for ettercap-graphical and click on it. Then enter your password for the device.
 - a. Note: If you are not using Kali and another distro use the command “sudo apt-get install -y ettercap-graphical” in order to install Ettercap.



3. After the Ettercap application is open, select the interface that you are using for example eth0, and then click the check button at the top as highlighted in the image above.



4. Then, Click the search icon in the top left and wait for the application to scan the subnet for devices, and then the application will provide a list of devices. After the scan is complete add the victim's IP address to target one person. Next, open Wireshark and select the same interface that was used earlier



Go back to ettercap and find the attacking tab at the top and click on the arp poisoning and click ok

Go back to wireshark and you should be able to go on the victim's and search for something and see it on the attackers' pc.

To stop the attack, you can click on the stop button at the top next to the attack tab.

To mitigate the ARP Spoofing attack, it is fairly simple. You first need to get into your switch. Then enter these basic commands in.

Enable DHCP Snooping:

```
Switch(config)# ip dhcp snooping
```

```
Switch(config)# ip dhcp snooping vlan 1
```

Then set the port connected to DHCP server/router to trust:

```
Switch(config)# interface fa0/24
```

```
Switch(config-if)# ip dhcp snooping trust
```

Enable DAI:

```
Switch(config)# ip arp inspection vlan 1
```

Then set routers and other switches that are currently on the network to trust: Switch(config)# interface fa0/24

```
Switch(config-if)# ip arp inspection trust
```

Problems:

I tried to use a virtual machine setup to simulate the attacker and victim on a single computer and that was successful for both the MAC flooding attack and the ARP spoofing attack, but when I had to use a different computer to simulate the DHCP starvation attack we were not able to run the attack due to multiple reasons. First, the computer itself could not properly run the Virtual Machine due to issues with the Ethernet connection to the other network devices which I was only to solve after we came up with another option. The reason was that on the back of the computer I had an Ethernet cable connected to an extension plug that connected to another Ethernet cable, but the extension plug was loose so the cables would randomly disconnect and reconnect causing the issue of the computer being booted off and on the network at random intervals. Another issue I had was the fact that some of the switches were running older software causing issues with certain messages and configurations which I was able to fix after completing this guide by comparing all of the configurations on each switch. After these problems occurred that I thought were associated with the virtual machine I decided instead of using a virtual machine, I would partition my drive on the computer so that the computer can either run Windows or Linux. I specifically used Kali Linux to run the DHCP starvation attack and were able to finish the attack successfully.

Conclusion:

In this analysis/guide, I demonstrated the 3 methods of how to be able to hack into a network and manipulate another user's device as well as how to mitigate these attacks

from happening in the first place. The 3 methods I explained were the DHCP Starvation attack, MAC Flooding attack and an ARP Spoofing attack.